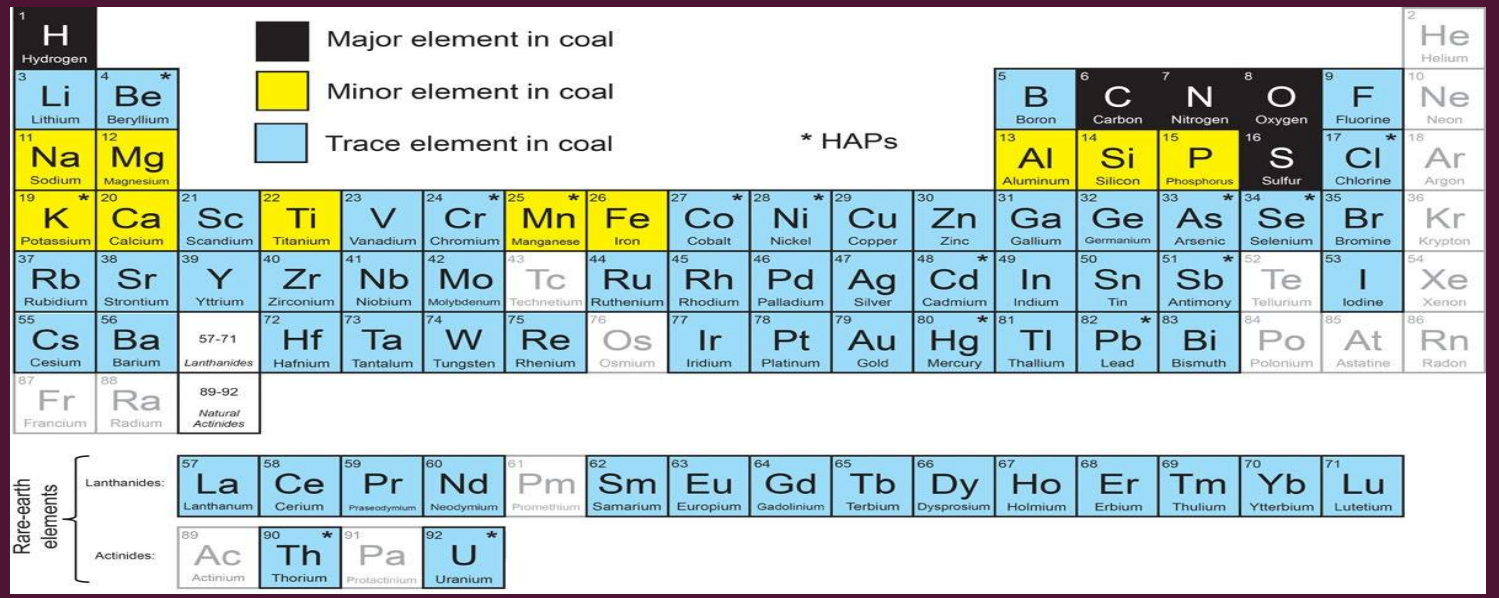
TRACE ELEMENTS

DR. HUMAIRA AMAN ALI



COPPER (CU)

- Total body copper is about 100 mg.
- It is seen in muscles, liver, bone marrow, blood, brain, kidney, heart and in hair.
- It occurs as:
  1. Erythrocuprein (in red blood cells as colorless)
  2. Hepatocuprein (in liver)
  3. Cerebrocuprein (in brain).

COPPER CONTAINING ENZYMES

- **Ceruloplasmin**
- **Cytochrome oxidase**
- **Cytochrome c**
- **ALA synthase**
- **Monoamine oxidase**
- **Superoxide dismutase.**

- Copper requirement for an adult is 1.5-3 mg per day.
- SOURCES:
  - Major dietary sources are cereals, meat, liver, nuts and green leafy vegetables.
  - Milk is very poor in copper content.
- Absorption:
  - Primarily absorbed from the duodenum.
  - Only about 10% of dietary copper is absorbed.
  - Phytates, Zinc, high amount of Vit. C inhibit Cu absorption.
- Excretion is mainly in feces via bile. Urine does not contain copper in normal circumstances.

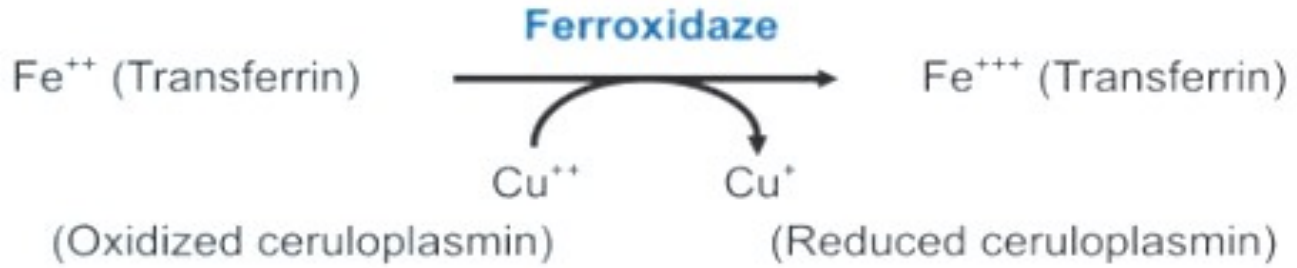
SERUM COPPER

Serum Cu is present in two distinct forms:

1. Transport form of Cu (4%): Which is loosely bound to albumin.

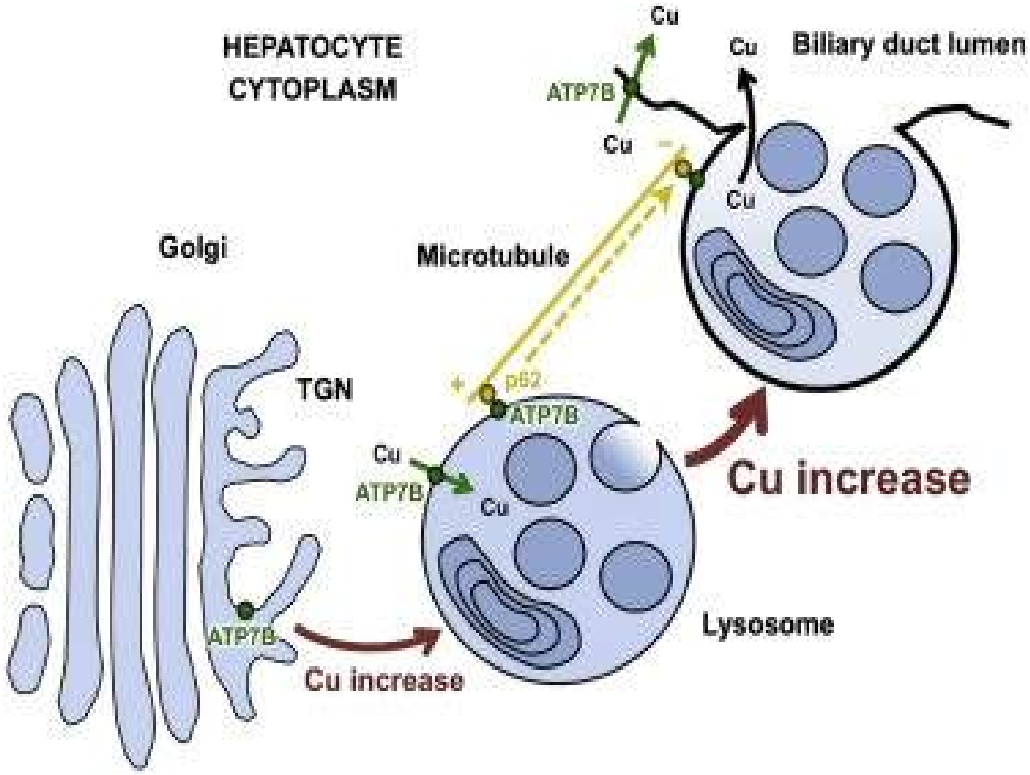
2. Bound form (96%): Which remains bound to α-globulin fraction of the serum, called "Caeruloplasmin" .

- In plasma ceruloplasmin is an important copper containing protein.
- Ceruloplasmin is a blue-colored glycoprotein.
- It is also called serum ferroxidase. It promotes oxidation of ferrous ion to ferric form, which is incorporated into trans



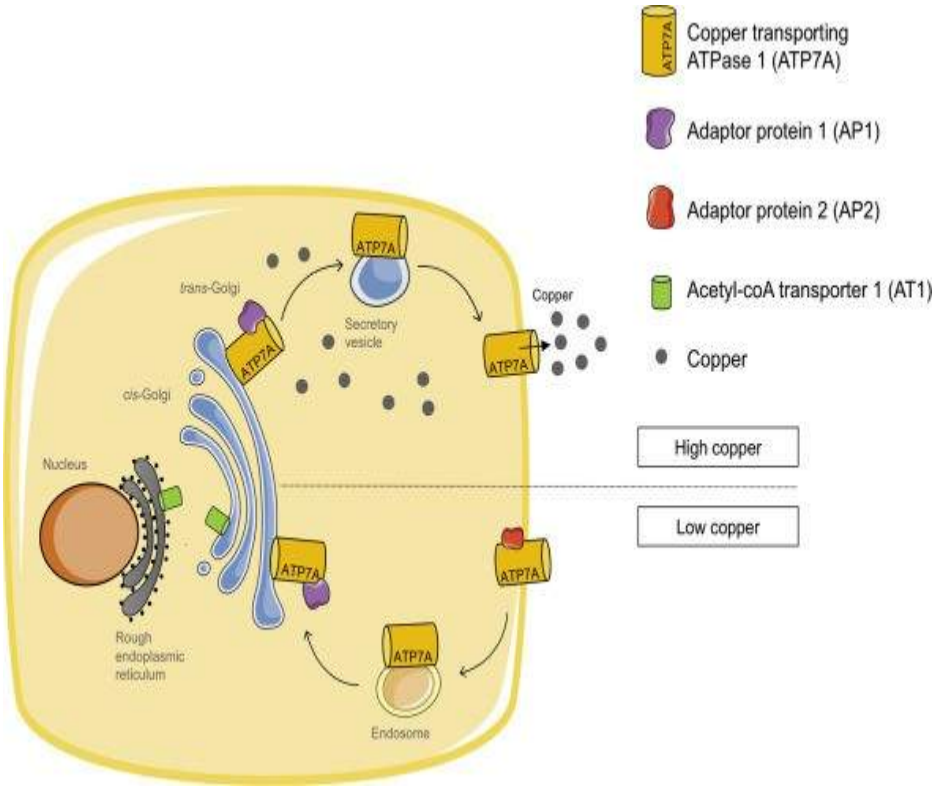
WILSON'S DISEASE (HEPATOLENTICULAR DEGENERATION):

- It is inherited as autosomal recessive. The incidence of Wilson's disease is 1 in 50,000.
- The basic defect is in a gene encoding a copper binding ATPase in cells (ATP7B gene in liver cells).
- This is required for normal transportation of copper from liver cells; in its absence, copper is accumulated in cells, leading to copper deposits in liver.

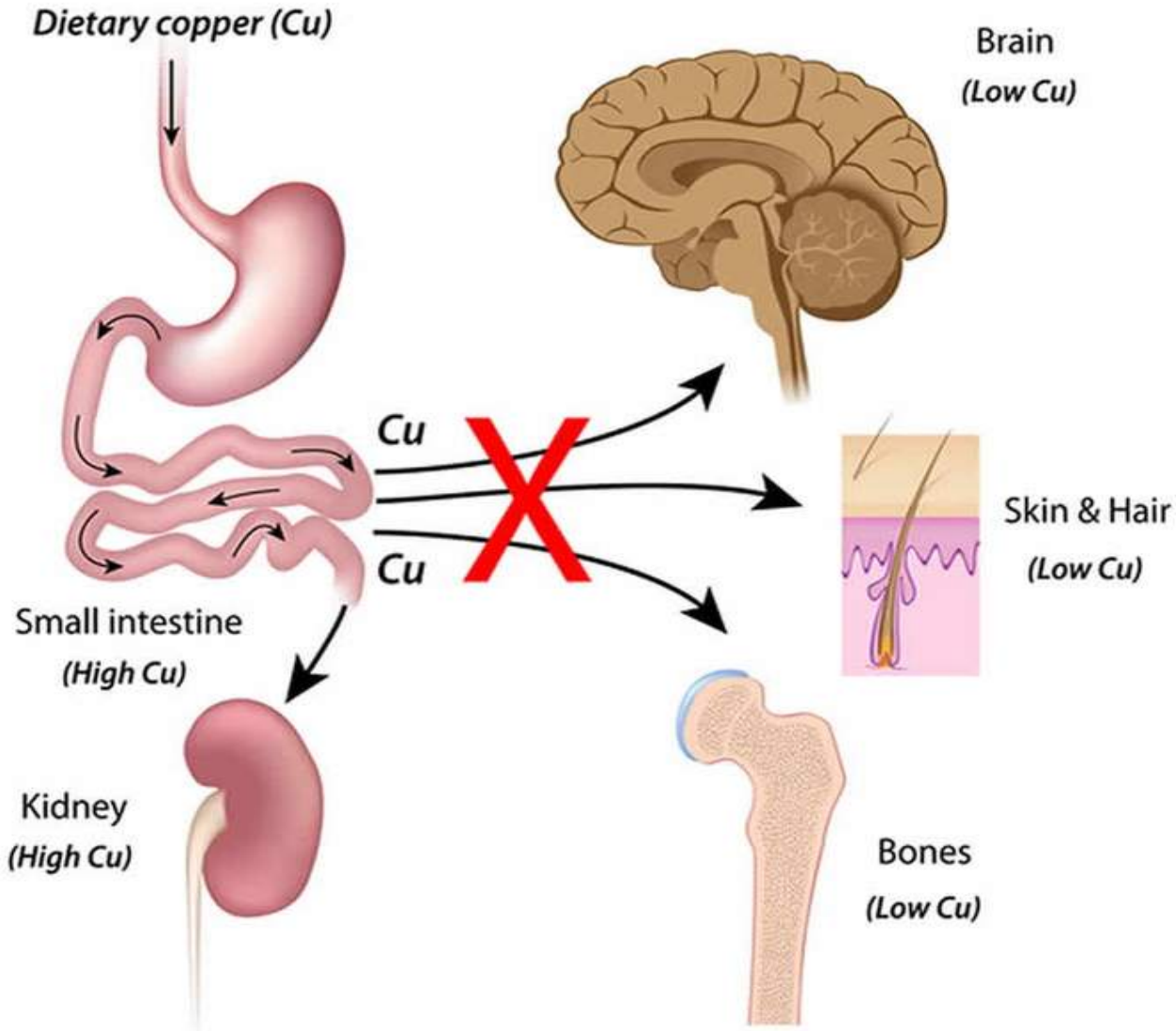


MENKE'S KINKY HAIR SYNDROME:

- It is an X-linked defect (affects only male children).
- It is a condition in which dietary copper is absorbed from GI tract; but cannot be transported to blood due to absence of an intracellular copper binding ATPase (mutation in ATP7A gene).
- Copper is not mobilized from intestinal cells as well as in other tissue cells.
- The copper that has entered into the cell is not able to get out of the cells, and so it accumulates.
- Copper deficiency thus leads to hypopigmentation and in extreme cases, grey color of hair.
- Low levels can cause brain dysfunction, especially of the cerebellum, leading to ataxia.



Menkes Disease





COPPER DEFICIENCY ANEMIA:

- Copper is essential for the formation of hemoglobin.
- Copper containing ceruloplasmin helps in iron transport.
- Copper is an integral part of ALA synthase, which is the key enzyme in heme synthesis.
- Copper helps the uptake of iron by normoblasts.
- Copper deficiency is manifested as anemia. RBC count is reduced; cell size is small.
- Copper deficiency thus results in microcytic normochromic anemia. If there is added iron deficiency, hypochromic anemia results

COPPER TOXICITY

- Chronic toxicity is manifested as diarrhea and blue-green discoloration of saliva.
- Copper poisoning may result in hemolysis, hemoglobinuria, proteinuria and renal failure.
- High levels have been associated with diseases like Alzheimer’s disease.

ZINC (ZN)

- Total zinc content of body is about 2 gm.

- Distribution:
  - High: In skin, and prostate
  - Average: In bones and teeth
  - Low: In kidneys, muscles, heart, pancreas and spleen and
  - Very low : In brain and lungs

REQUIREMENT OF ZINC

- For adults is 10 mg/day
- Children 10 mg/day
- In pregnancy and lactation 15-20 mg/day
- Since iron inhibits absorption of zinc, when iron is supplemented, zinc is also given to prevent any deficiency.

SOURCES, ABSORPTION & EXCRETION

- Sources
  - Animal sources: liver, milk and dairy products, eggs.
  - Vegetable sources: legumes, pulses, oil seeds, yeast cells, and vegetables (spinach, lettuce)
- Copper, calcium, iron and phytate will interfere with the absorption of zinc.
- Zinc and copper will competitively inhibit each other's absorption. So zinc is therapeutically useful to reduce copper absorption in Wilson's disease.
- Only a small percentage of dietary zinc is absorbed.
- Zinc is excreted through faeces and to a lesser extent through urine. Trace amount is lost in sweat.

ROLE IN ENZYME ACTION:

- Zinc forms an integral part of several enzymes (metallo-enzymes) in the body.

- Important zinc containing enzymes are:
  - Carbonic anhydrase, alkaline phosphatase, lactate dehydrogenase, alcohol dehydrogenase and glutamate dehydrogenase.
  - RNA polymerase contains zinc and so it is required for protein biosynthesis.
  - Superoxide dismutase is zinc dependent and so, zinc has antioxidant activity.

OTHER FUNCTIONS

- Insulin when stored in the beta cells of pancreas contains zinc, which stabilizes the hormone molecule thus participate in storage and secretion of Insulin.
- Zinc containing protein in saliva is important for taste sensation.
- Role in Vitamin A Metabolism: Zn++ has been claimed to stimulate the release of vitamin A from liver into the blood.
- Zinc is necessary for wound healing. Zinc deficiency delays wound healing.
- Role in Growth and Reproduction: zinc deficiency may lead to ‘dwarfism’ and ‘hypogonadism’. Zinc deficiency also lowers spermatogenesis in males and menstrual cycles are disturbed in females

ZINC DEFICIENCY MANIFESTATIONS

- Poor wound healing
- lesions of skin
- Impaired spermatogenesis
- Hyperkeratosis
- Dermatitis and alopecia
- There is reduction in number of T and B lymphocytes.
- Macrophage function is retarded
- Zinc deficiency leads to depression, dementia and other psychiatric disorders

ACRODERMATITIS ENTEROPATHICA

- It is a recessive condition where zinc absorption is defective and is characterized by:
- Acrodermatitis (inflammation around mouth, nose, fingers, etc.)
- Diarrhea
- Alopecia (loss of hair in discrete areas)
- Ophthalmoplegia and
- Hypogonadism.



FLUORIDE (F^-)

- Fluoride is the negative ion of the element fluorine.
- It's widely found in nature, in trace amounts.
- It occurs naturally in air, soil, plants, rocks, fresh water, seawater, and many foods.

Sources:

- Fluoride is derived in human from drinking water.
- Other sources: Tea and salmon contain small amounts of fluoride.

REQUIREMENTS:

- About one part of fluorine in one million parts of drinking water (one PPM) seems to serve the daily requirement of fluorine in human adults and children.
- Daily intake of fluoride should not exceed 3 mg as it is a toxic element.
- For an adult individual, the lethal dose is 2.5 gm.

FUNCTIONS

- Fluoride is present in trace amounts in all mineralized tissues of the body such as enamel, dentin, and bone and plays an important role in the mineralization of bones and teeth, helps in tooth development, normal maintenance and hardening of dental enamel and prevention of “dental caries”.

- In mineralized tissues and biomaterials, fluoride ions increase the stability of mineralized tissues and materials by decreasing the solubility of hydroxy-apatite mineral.

- The protective effects of fluoride on dental health were first observed in 1930 as there was less tooth decay in communities consuming naturally fluoridated water compared to non-fluoridated areas.

- Due to these beneficial effects of fluoride, it was introduced into dentistry in 1940 and since then, it is being added to various consumer products.

- Greater than 1.2 PPM of fluorine in drinking water of infants/children may increase fluoride contents of the enamel and dentine, may reduce Ca deposition in those tissues and may cause mottling of enamel, in newly erupted permanent teeth: Discoloration, corrosion and stratification of enamel.
- It also promotes normal bone development, increases retention of Ca++ and PO4 and prevent old age osteoporosis.
- High fluoride intakes may raise the fluoride content of bone, stimulate osteoblast activity and cause an abnormal rise in calcium deposition and increased density of bone.

FLUORIDE TOXICITY "FLUOROSIS":

- Excess of fluoride in water or diet is harmful and is considered to be the main cause of the crippling disease known as 'fluorosis'.
- The fluoride levels in GI tract, blood and urine have been reported to be high in fluoride toxicity.
- High fluoride concentration inhibits particularly Mg++ dependent enzymes. The structural and biochemical changes led to muscle wasting and impairment of energy metabolism.
- 10 to 46% reduction in protein contents of various organs, e.g. adrenal gland, cardiac muscles, kidneys, lungs, pancreas, skeletal muscles, spleen, stomach, testes and spinal cord, suggestive of inhibition of protein synthesis by fluorides.

cont..

- Steroid production was impaired because of depletion of β-hydroxysteroid dehydrogenase enzyme.

- Collagen content is found to be reduced and its biosynthesis adversely affected. Formation of deficient collagen fibres with abnormal biochemical sites, provide an impetus for pathological calcification which occurs during fluoride intoxication.

IODINE

- Daily requirement of iodine is 150-200 micrograms/day.
- Its sources are drinking water, fish, cereals, vegetables and iodinated salt.
- Total body contains 25-30 mg of iodine.
- All cells do contain iodine; but 80% of the total is stored in the thyroid gland.
- Upper regions of mountains generally contain less iodine. Such areas are called goiterous belts.

cont..

- The program of iodination of common salt has resulted in increased availability of iodine.
- Ingredients in foodstuffs, which prevent utilisation of iodine are called goitrogens. Goitrogens are seen in maize, millet, bamboo shoots, sweet potatoes and beans. Cabbage contain thiocyanate, which inhibits iodine uptake by thyroid.
- The only biological role of iodine is in formation of thyroid hormones, thyroxine (T4) and tri-iodo thyronine (T3).

MANGANESE

- Manganese is also an essential trace element and is required by the body.
- Acts either as a ‘cofactor’ or as an activator of many enzymes like arginase, isocitrate dehydrogenase (ICD), cholinesterase, lipoprotein lipase, enolase, leucine amino peptidase in intestine, phosphotransferases and 5-oxo-prolinase of kidneys, and small intestine and many others.
- Mn++ plays a part in the synthesis or deposition of Mucopolysaccharides (MPS) in the cartilaginous matrices of long bones.

- Mn-deficiency causes significant lowering in the content of chondroitin SO4.
- Manganese also helps in porphyrin synthesis by participating in δ-ALA synthetase activity.
- Hb-synthesis appears to be depressed in Mn-deficient rats.
- Manganese has been reported also to exhibit “lipotropic effect”(removal of fat from liver) and it stimulates FA synthesis and cholesterol synthesis.
- Manganese also participates in glycoprotein and proteoglycan synthesis

SELENIUM (SE)

- It is widely distributed in all the tissues, highest concentrations are found in liver, kidneys and fingernails. Muscles, bones, blood and adipose tissues show a low concentration of selenium.
- Requirement is 50-100 microgram/day. Normal serum level is 50-100 microgram/dl.
- Glutathione peroxidase (GP) is the important selenium containing enzyme.
- RBC contains good quantity of glutathione peroxidase.
- Thyroxin is converted to T3 by 5'-de-iodinase, which is a selenium containing enzyme.

- In Se deficiency, this enzyme becomes less active, leading to hypothyroidism.
- Selenium concentration in testis is the highest in adult tissue. It is necessary for normal development of spermatozoa.
- It is supplementary to vitamin E and acts as primary antioxidant by scavenging reactive oxygen species and free radical intermediates of polyunsaturated lipid peroxidation.

COBALT

- Cobalt forms an integral part of vitamin B12 and is required as a constituent of this vitamin.
- Cobalt is required to maintain normal bone marrow function and required for development and maturation of red blood cells.
- A deficiency of cobalt results in decreased B12 supply which produces nutritional macrocytic anaemia.
- Cobalt may act as a cofactor for enzyme like glycyl-glycine dipeptidase of intestinal juice.

NICKEL

- Nickel activates several enzyme systems, viz. arginase, carboxylase, trypsin and acetyl-CoA synthetase.

- Nickel is required in trace amount for growth and reproduction.

THANK YOU